

REVISED ENGINEERING PROPOSAL

NeoNodes

Complete GPU Marketplace

A fully automated, standalone GPU compute marketplace. From node runner onboarding to customer job submission, automatic execution, yield optimization, and settlement — with zero manual intervention required.

PROJECT
NeoNodes Platform

CLIENT
Emmanuel

DATE
24 March 2026

VERSION
2.0 — Full Platform

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Executive Summary

NeoNodes is a complete, standalone GPU compute marketplace that replaces the current manual TokenOS workflow. Node runners onboard themselves, customers submit jobs directly, and the A²E engine handles all routing, execution, and settlement automatically.

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GPU TIERS

3

MARKETS

150+

FEATURES

100%

AUTOMATED

What This Platform Delivers

✓

Self-Service Node Onboarding
Node runners register, download agent, and start earning automatically

✓

Customer Job Submission
Customers submit compute jobs via portal or API with instant pricing

✓

Automatic Job Execution
Node agent receives jobs and executes Docker containers autonomously

✓

A²E Yield Optimization
Routes to highest-paying market with cost floor protection

✓

External Market Integration
Overflow to Akash & IO.net when internal demand is low

✓

Automated Settlement
Track earnings, calculate payouts, process settlements on schedule

Zero Manual Intervention

Once deployed, NeoNodes operates fully autonomously. Nodes join, jobs execute, payments settle — all without admin involvement. Your time is freed from manual compute management entirely.

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Progress Update

Work completed to date under the original Phase 1 agreement. The foundation is built and operational — ready for expansion to full standalone platform.

Milestone 1 — Architecture & Design

\$800 COMPLETE

DELIVERED

- Full system architecture document with service diagrams
- Database schema design — 12 tables with Prisma ORM
- OpenAPI specification for all REST endpoints
- Pricing model and arbitrage logic specification
- Provider yield floor model with worked examples
- Job abstraction and failure scenario documentation
- Arbitrage rule set design with decision flow
- Monorepo structure with shared TypeScript types

Milestone 2 — Node Registry, Orchestration & A²E Engine

\$1,100 COMPLETE

DELIVERED

- Node Registry API — registration, listing, details, deregistration
- Heartbeat system — 30-second health checks with GPU metrics
- Node health state machine — ONLINE → DEGRADED → OFFLINE
- A²E Routing Engine — compares Internal vs Akash vs IO.net rates
- Rule-based routing with yield floor enforcement
- BullMQ job queue with retry logic (3 attempts, exponential backoff)
- Auto node assignment with load balancing
- Full job lifecycle tracking — PENDING → ASSIGNED → COMPLETED
- Rate fetching from Akash (live) and IO.net (mock)
- Redis caching with 60-second rate refresh
- WebSocket server for real-time events
- Configuration API — yield floors, market toggles
- Statistics API — nodes, jobs, earnings aggregations
- API authentication with X-API-Key validation

COMPLETED INVESTMENT

\$1,900

M1 + M2 delivered and operational

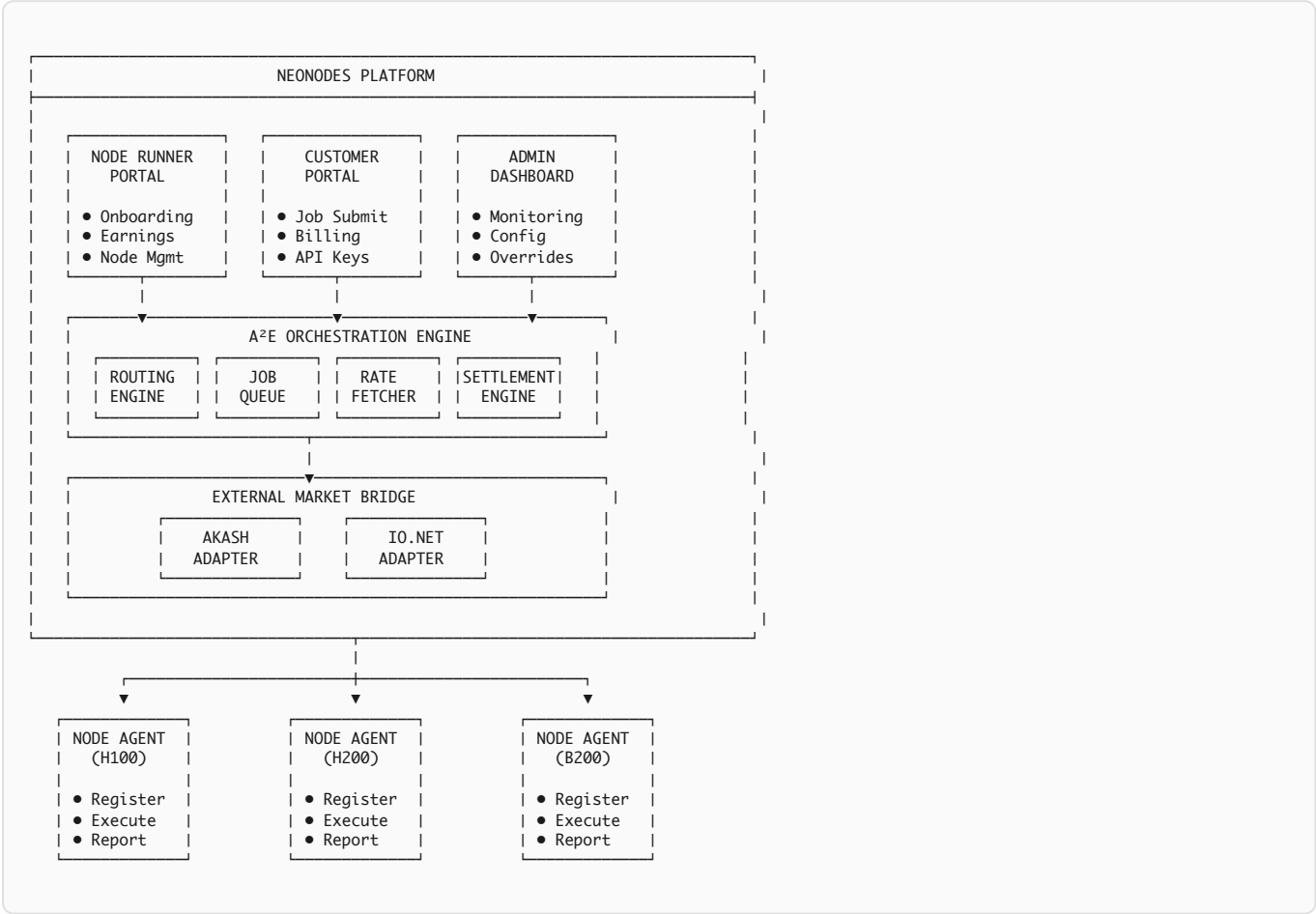
API STATUS

LIVE<https://a2e.byredstone.com>

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Platform Architecture

Complete system architecture for the fully automated NeoNodes marketplace.



Data Flow

STEP	COMPONENT	ACTION
1	Node Runner Portal	Node runner signs up, connects wallet, downloads agent
2	Node Agent	Agent registers with A²E, starts sending heartbeats
3	Customer Portal	Customer submits job with GPU requirements
4	A²E Routing Engine	Compares rates, selects market, assigns to best node
5	Node Agent	Receives job, pulls Docker image, executes workload
6	Node Agent	Reports completion with duration and metrics
7	Settlement Engine	Calculates earnings, queues payout to node runner

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Scope of Work

Complete breakdown of all remaining features required to deliver the fully automated NeoNodes marketplace platform.

Module A — Admin Dashboard	PENDING
DASHBOARD CORE • Next.js 14 application with App Router architecture • Admin authentication with secure session management • Responsive design optimized for desktop and tablet • TokenOS-style dark theme with real-time WebSocket updates OVERVIEW DASHBOARD • Key metrics cards — total nodes, active jobs, earnings today • Live activity feed with real-time event streaming • System health indicators — API, database, Redis, external markets • Earnings visualization with time-series charts • Market status comparison — Internal vs Akash vs IO.net NODE MANAGEMENT UI • Sortable, filterable table of all registered nodes • Node detail view — specs, metrics history, job history • Node status controls — pause, resume, maintenance mode • GPU health visualization — temperature, utilization charts • Bulk operations for multi-node actions JOB MANAGEMENT UI • Job list with status, market, node, earnings columns • Job detail view — full info, routing decision, timeline • Job actions — cancel, retry, manual reassignment • Visual lifecycle progression indicator • Advanced filtering by status, market, node, date range CONFIGURATION UI • Yield floor editor — adjust per GPU tier with instant preview • Market toggle controls — enable/disable external markets • Rate override capability for manual adjustments • Configuration change audit log ROUTING SIMULATOR • Test interface to simulate routing decisions • What-if analysis for yield floor changes • Side-by-side market rate comparison	

Module B — Financial System

PENDING

EARNINGS TRACKING

- Automatic per-job earnings calculation on completion
- Per-node earnings aggregation with historical tracking
- Per-market revenue breakdown — Internal, Akash, IO.net
- Per-GPU-tier earnings analysis
- Automated daily earnings rollup and summaries
- Earnings trend analysis with visualizations

COST TRACKING

- Infrastructure cost recording and tracking
- External market deployment cost tracking (Akash/IO.net)
- True cost-per-job calculation
- Margin analysis — revenue minus costs per market
- Profitability reporting by tier and market

SETTLEMENT ENGINE

- Payout calculation per node runner based on completed jobs
- Configurable settlement periods — daily, weekly, monthly
- Settlement queue with batch processing
- Settlement status tracking — pending, processing, completed, failed
- Minimum threshold configuration before payout triggers
- Settlement retry logic for failed payouts

PAYMENT PROCESSING

- Solana wallet integration for SOL/USDC payouts
- Efficient multi-recipient payout batching
- Full transaction recording with blockchain references
- On-chain payment verification and confirmation
- Failed payment handling with automatic retry

REPORTING

- CSV export — earnings, settlements, jobs, nodes
- PDF invoice generation for customer billing
- Monthly earning statements for node runners
- Tax-ready annual summaries for compliance
- Custom date range reporting with flexible filters

FINANCIAL DASHBOARD

- Revenue overview with trends and projections
- Cost overview with expense breakdown
- Net profit margins by market, tier, and time period
- Pending settlements visibility and management

Module C — Node Agent

[NEW SCOPE](#)

AGENT CORE

- Standalone binary executable for easy deployment
- Automatic registration with A²E on first startup
- Secure TLS-encrypted API communication
- Heartbeat system sending health metrics every 30 seconds
- Graceful shutdown — complete current job before stopping
- Self-update capability to latest agent version
- Configuration file with customizable settings

GPU MANAGEMENT

- Automatic GPU detection via nvidia-smi integration
- Multi-GPU support — handle multiple GPUs per node
- Real-time GPU metrics — temperature, utilization, memory usage
- GPU allocation — assign specific GPU to specific job
- Health monitoring — detect GPU failures and thermal throttling
- VRAM tracking and availability reporting

JOB EXECUTION

- Job polling — check for assigned jobs on interval
- Job acceptance — confirm job receipt to A²E
- Docker integration — pull images, manage containers
- Resource limits — enforce CPU, memory, GPU constraints
- Execution timeout — kill jobs exceeding time limit
- Output capture — collect stdout/stderr logs
- Environment variable injection for job configuration
- Volume mounting for data persistence

STATUS REPORTING

- Job started notification — report RUNNING status instantly
- Progress updates — periodic progress reporting to A²E
- Job completed — report success with duration and metrics
- Job failed — report failure with error details and logs
- Resource usage metrics during execution

FAILURE RECOVERY

- Crash recovery — resume state after agent restart
- Job checkpointing — save state for long-running jobs
- Automatic reconnection with exponential backoff

SECURITY

- Container isolation — sandboxed execution environment
- Resource limits — prevent resource exhaustion attacks
- Network policies — restrict container network access
- Image verification — only run trusted, verified images
- Secure credential storage for API authentication

INSTALLATION

- One-line installer script — curl | bash deployment
- Docker prerequisite check and guidance
- NVIDIA driver verification, guided setup wizard, systemd auto-start

Module D — Node Runner Portal

[NEW SCOPE](#)

AUTHENTICATION

- Wallet Connect integration — Phantom, Solflare support
- Optional email registration for notifications
- Secure session management with JWT tokens
- Optional two-factor authentication (2FA)
- Password recovery flow for email-based accounts

ONBOARDING FLOW

- Welcome wizard with step-by-step guidance
- System requirements display — GPU specs, Docker, drivers
- Platform-specific agent download — Linux, Windows, macOS
- Detailed installation guide with screenshots
- Automatic verification when node comes online
- Optional GPU benchmark for performance baseline
- Terms of service and agreement acceptance

DASHBOARD

- Earnings overview — today, this week, this month, all-time
- Node status display — online/offline indicator, current job
- Performance metrics — uptime percentage, jobs completed, efficiency
- Earnings chart with time-series visualization
- Recent activity feed showing job completions

NODE MANAGEMENT

- Node list showing all registered nodes
- Node details — specifications, status, job history
- Pause/Resume controls — temporarily stop accepting jobs
- Maintenance mode scheduling for planned downtime
- Node removal — deregister from platform
- Multiple node support per account

EARNINGS & PAYOUTS

- Complete earnings history with transaction details
- Pending earnings display — unsettled amounts
- Payout history — all completed payouts with blockchain links
- Payout settings — minimum threshold, frequency preferences
- Wallet management — update payout wallet address
- Earnings projection based on historical performance

JOB HISTORY

- List of all completed jobs with details
- Job detail view — duration, earnings, market routed
- Execution logs access (if enabled by admin)
- Filtering and search by date, status, earnings

NOTIFICATIONS

- Email alerts — node offline, payout sent, important updates
- In-app notification feed with real-time updates
- Notification preferences — configure what to receive
- Browser push notifications (optional)

Module E — Customer Portal

[NEW SCOPE](#)

AUTHENTICATION

- Wallet Connect integration — Phantom, Solflare
- Traditional email/password authentication option
- API key generation for programmatic access
- Team management — multiple users per organization
- Role-based access control (admin, developer, viewer)

JOB SUBMISSION

- Job creation form with GPU tier selection
- Container configuration — image, command, environment
- Resource selection — GPU count, memory, storage
- Cost estimation displayed before submission
- Job templates — save and reuse configurations
- Batch job submission for multiple jobs
- Priority levels for urgent jobs

JOB MONITORING

- Active jobs list with real-time status updates
- Job status progression — queued, running, completed
- Live log streaming during execution
- Resource metrics — GPU/CPU usage during execution
- Cancel job functionality
- Job queue position visibility

JOB HISTORY

- Historical job list with search and filters
- Job details — full execution information
- Output download — retrieve job outputs
- Rerun job — clone configuration and resubmit
- Execution logs archive

BILLING

- Account balance display
- Add funds — SOL/USDC deposit with QR code
- Usage history with detailed breakdown
- Invoice download — PDF invoices
- Spending limits — set maximum spend thresholds
- Auto-recharge configuration

API ACCESS

- Interactive API documentation with examples
- API key management — create, revoke, rotate
- API usage monitoring and rate limit visibility
- SDK downloads — Python, JavaScript client libraries
- Webhook configuration for job status callbacks

Module F — External Market Execution

[NEW SCOPE](#)

AKASH NETWORK INTEGRATION

- Akash wallet setup and AKT balance management
- SDL (Stack Definition Language) generation for jobs
- Deployment creation via Akash CLI/API
- Automatic bid selection from providers
- Deployment status monitoring and health checks
- Log retrieval from Akash deployments
- Clean deployment termination on completion
- AKT cost tracking per deployment
- Provider reputation scoring

IO.NET INTEGRATION

- IO.net API authentication and account management
- Cluster creation and configuration
- Job submission to IO.net network
- Real-time status monitoring
- Result retrieval and output handling
- Cost tracking in IO credits
- Failover handling between providers

MARKET ABSTRACTION

- Unified interface across all external markets
- Automatic failover — switch markets on failure
- Cost normalization — compare apples to apples
- Result aggregation — unified job results format
- Market health monitoring and availability tracking
- Rate limiting and quota management per market

BIDIRECTIONAL ROUTING

- Internal demand high → serve internal customers at premium rates
- Internal demand low → list idle nodes on Akash/IO.net for external jobs
- Automatic switching based on demand levels
- Node runners always earning — internal OR external, never idle
- Hourly rate calculation regardless of job source

Bidirectional Market Routing

High Internal Demand: Node runners serve internal jobs at premium rates.

Low Internal Demand: Idle nodes automatically list on Akash/IO.net to earn from external demand.

Result: Node runners always earn — either from internal customers or external markets, calculated hourly.

Future: Data Center API Automation

Node deployment currently requires manual data center coordination. If Hydrahost or similar providers offer API-based provisioning, automated node deployment can be added as a future enhancement at no additional cost.

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Tech Stack

100% Node.js — one language across the entire platform. Fast, unified, and fully owned.

Node.js

Fastify

Next.js 14

React

Vite

Tailwind CSS

Socket.io

BullMQ

Redis

PostgreSQL

Prisma

Docker

Akash SDK

IO.net API

Solana Web3.js

LAYER	TECHNOLOGY	PURPOSE
Backend API	Node.js + Fastify	Core A²E engine, node registry, job processing, settlement
Admin Dashboard	Next.js 14 + React	Full admin control — monitoring, configuration, overrides
Node Runner Portal	Next.js 14 + Tailwind	Self-service onboarding, earnings tracking, node management
Customer Portal	Next.js 14 + Tailwind	Job submission, billing, API access, monitoring
Node Agent	Node.js + Docker SDK	GPU node daemon — job execution, metrics, heartbeat
Real-time	Socket.io	Live updates across all dashboards and portals
Job Queue	BullMQ + Redis	Async processing, retry logic, rate limiting
Database	PostgreSQL + Prisma	Node registry, jobs, earnings, settlements, audit logs
Payments	Solana Web3.js	SOL/USDC payouts to node runners
External Markets	Akash SDK + IO.net API	Live rates, job deployment, status monitoring

Why Node.js for Everything?

One language from backend to frontend to node agent means faster development, easier debugging, shared types and utilities, and no context switching. The entire team works in TypeScript. No Python, no Go, no fragmentation — just JavaScript everywhere.

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Milestone Breakdown

Seven milestones to deliver the complete NeoNodes platform. Payment released after client review and approval of each milestone.

MILESTONE	SCOPE	STATUS	INVESTMENT
M1	Architecture & Economic Design	✓ COMPLETE	\$800
M2	Node Registry, Orchestration & A²E Engine	✓ COMPLETE	\$1,100
M3	Admin Dashboard & Configuration UI	PENDING	\$1,200
M4	Financial System & Settlement Engine	PENDING	\$1,600
M5	Node Agent & Execution Runtime	NEW	\$2,000
M6	Node Deployer Portal & Compute Buyer API	NEW	\$2,100
M7	External Market Overflow (Akash, IO.net, Vast.ai)	NEW	\$1,200

TOTAL PLATFORM INVESTMENT

\$10,000

~~\$11,200~~ DISCOUNTED • \$1,900 paid • \$8,100 remaining

TIMELINE

9 Weeks

From M3 start to full platform

Payment Schedule

MILESTONE	PAYMENT	TRIGGER
M1 + M2	\$1,900	✓ PAID
M3 — Admin Dashboard	\$1,200	Dashboard delivered and reviewed
M4 — Financial System	\$1,600	Settlement engine operational
M5 — Node Agent	\$2,000	Agent executing jobs autonomously
M6 — Portals	\$2,100	Node runner and customer portals live
M7 — External Markets	\$1,200	Akash & IO.net execution working

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Delivery Timeline

Estimated delivery schedule for remaining milestones. Actual timing may vary based on feedback cycles and scope refinements.

MILESTONE	DURATION	CUMULATIVE	KEY DELIVERABLES
M3	1 week	Week 1	Admin dashboard with full monitoring and controls
M4	1.5 weeks	Week 2.5	Earnings tracking, settlement engine, reports
M5	2 weeks	Week 4.5	Node agent with Docker execution runtime
M6	2.5 weeks	Week 7	Node runner portal + customer portal
M7	2 weeks	Week 9	Akash & IO.net job execution

Progressive Deployment

Each milestone delivers working functionality. You don't wait 9 weeks to see results — after M3 you have a dashboard, after M4 you have settlement, after M5 you have automatic execution. The platform becomes more capable with each milestone.

What You Get at Each Stage

After M3 + M4

- ✓ Full admin visibility and control
- ✓ Earnings tracking and settlement
- ✓ Manual job submission via API
- ✓ Routing decisions logged

After M7 (Complete)

- ✓ External market overflow
- ✓ 100% GPU utilization
- ✓ Multi-market arbitrage
- ✓ Full NeoNodes platform

After M5 + M6

- ✓ Node runners self-onboard
- ✓ Jobs execute automatically
- ✓ Customers submit via portal
- ✓ Zero manual intervention

Business Impact

- ✓ No more manual node management
- ✓ Scale without headcount
- ✓ Maximize node runner yield
- ✓ Compete with Akash & IO.net

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Ready to Continue

The foundation is built and operational. We're ready to proceed with the remaining milestones to deliver the complete NeoNodes platform.

**API Live**
A²E engine operational at <https://a2e.byredstone.com>

**Database Ready**
PostgreSQL with full schema deployed

**Code Repository**
Private GitHub repo with full documentation

**Rate Fetching**
Akash rates updating every 60 seconds

Next Steps
Upon approval of this revised proposal, we will immediately begin M3 (Admin Dashboard). The dashboard will provide full visibility into the operational A²E engine while we build out the remaining platform components.

Investment Summary

\$1,900 COMPLETED	\$8,100 REMAINING	\$10,000 TOTAL (DISCOUNTED)
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Complete NeoNodes platform with automated node onboarding, customer portal, job execution, settlement, and external market integration.